



Muscle fiber types in mutton: implications for meat quality and flavor formation—a review



Mengjie Chen^{1,4}, Xiaofei Zhang^{2,4}, Nian Zhang^{1,3}, Aishao Shangguan¹, Fan Jiang¹, Xiaojun Suo¹, Jin Chai² & Qi Xiong¹✉

Mutton quality and flavor are closely associated with muscle fiber type composition. This review summarizes muscle fiber classification, mutton flavor formation pathways, and the effects of muscle fiber types on color, pH, tenderness, juiciness, and flavor compounds. It also discusses how genotype, age, sex, feeding practices, and nutrition regulate mutton quality through muscle fiber transformation, providing references for high-quality mutton production.

Mutton is tender and juicy, delicious and unique in flavor. It has the advantages of high protein, low fat and cholesterol, rich in amino acids, vitamins and trace elements necessary for the human body, which occupies an important position in people's diet structure and deeply loved by consumers¹. According to the data of the National Bureau of Statistics (NBS), in 2024, the total output of pork, beef, mutton and poultry in China was as high as 96.63 million tons, of which the annual output of pork was 57.06 million tons, the annual output of poultry was 26.6 million tons, the annual output of beef was 7.79 million tons, and the annual output of mutton was only 5.18 million tons, accounting for only 5.36% of the total output of pork, beef, mutton and poultry in China, which was far lower than that of pork, poultry and beef². There are many reasons for the low yield of mutton, among which the meat quality and special flavor of mutton directly affect the acceptance and market value of consumers. Research has found that meat tenderness, taste, and fat content are considered by the majority of consumers to be highly decisive factors. Elderly people consider tenderness more important than flavor, preferring cuts with softer, more tender meat, while younger individuals prefer cuts with richer flavors³. Consumers dissatisfaction with the flavor and odor of mutton is the primary constraint on the overall consumption quality, with strong odors and unpleasant tastes pose significant barriers to consuming mutton⁴. Foreign scholars describe this undesirable flavor characteristics of mutton as “animal odor,” “rotten odor,” “metallic odor,” “fecal odor,” or “camphor odor,” whereas in China, the term “gamey odor” is commonly used to characterize the distinctive flavor of mutton⁵. In recent years, in pace with the continuous improvement of people's requirements for food quality, the research on the flavor of mutton has been paid more and more attention. Improving mutton quality, enhancing flavor, and reducing mutton odor has become a core issue, which require urgent solutions for enhancing efficiency and ensuring sustainable

development in the mutton industry. Meat quality and flavor are the core indicators of sheep breeding, mainly manifested in color, pH value, tenderness, intramuscular fat (IMF), juiciness, and flavor. Their formation is a complex process, the result of the interaction of genetic, physiological, metabolic, and environmental factors. As the basic functional units of muscle tissue, muscle fibers are a crucial link connecting “physiological metabolism” with “meat quality and flavor.” Studies have shown that factors such as muscle fiber type, diameter, density and metabolic characteristics can affect the mutton tenderness, juiciness, and flavor compound content, ultimately shaping its overall sensory characteristics⁶. Therefore, elucidating the pathways of mutton flavor formation, analyzing the relationship between muscle fiber types and mutton quality and flavor, and delineating the factors that affect muscle fiber types, meat quality and flavor are of great significance for improving mutton quality, reducing odors, meeting consumer demands, and promoting the sustainable development of the sheep farming industry.

Muscle fiber characteristics and their distribution in mutton

Muscle fiber is the most fundamental structural unit of muscle, accounting for about 75–90% of muscle volume. It is elongated, multinucleated and spindle-shaped, composed of numerous densely packed myofibrils, which plays a crucial role in regulating the metabolism and endocrine functions in livestock and poultry⁷. Livestock muscles contain various muscle fiber types, and the differences in their contractile properties and metabolic patterns are key determinants of meat tenderness, juiciness, and flavor. The traditional classification method categorizes muscle fibers into three types: red muscle fibers, white muscle fibers, and intermediate muscle fibers between the two. Alternatively, they can also be broadly classified into fast-twitch and slow-

¹Institute of Animal Sciences and Veterinary Medicine, Key Laboratory of Animal Embryo Engineering and Molecular Breeding of Hubei Province, Hubei Academy of Agricultural Sciences, Wuhan, China. ²Key Laboratory of Agricultural Animal Genetics, Breeding, and Reproduction of Ministry of Education, Huazhong Agricultural University, Wuhan, China. ³Wuhan Luxiang Agricultural Development Co. Ltd, Wuhan, China. ⁴These authors contributed equally: Mengjie Chen, Xiaofei Zhang.

✉ e-mail: phenixxq@163.com

twitch fibers. In 1970, Brooke et al. divided muscle fibers into three types: type I, type IIA and type IIB muscle fibers according to the different myosin ATPase activity⁸. Jouffroy et al. used NADH-TR staining method to divide muscle fibers into four types: type I, type IIA, type IIB and type IIX⁹. With the advancement of molecular biology techniques, the classification of muscle fibers has increasingly centered on the myosin heavy chain (MyHC) isoforms as a core biomarker. These include MyHC 7, MyHC 4, MyHC 2, and MyHC 1, which correspond to type I, type IIA, type IIB, and type IIX muscle fibers, respectively¹⁰. These four types represent slow-contracting oxidized muscle fibers, fast-contracting oxidized glycolytic muscle fibers, fast-contracting glycolytic muscle fibers, and intermediate muscle fibers, respectively. In mutton, the composition and distribution of these fiber types show significant site-specificity and breed variation, which is an intrinsic reason for differences in texture and flavor between different cuts. Currently, the most studied muscles regarding mutton fiber characteristics are the longissimus dorsi (loin) and biceps femoris (round). Study reported the muscle fiber type composition of the longissimus dorsi and biceps femoris muscle in Sunite sheep, revealing that the proportions of type I, IIA, and IIB fibers in the longissimus dorsi were 8.74%, 23.36%, and 67.90%, respectively, while those in the biceps femoris muscle were 16.34%, 29.96%, and 53.69%, respectively. The proportion of muscle fiber types in the two parts is as follows: type IIB > type IIA > type I, with type IIB fibers being the main type¹¹. In black Tibetan sheep, the proportions of type I muscle fibers in the longissimus dorsi, triceps brachii, and biceps femoris were 12.55%, 12.34%, and 11.92%, respectively. The proportion of type IIA muscle fibers is 32.26%, 29.79%, and 25.71%, respectively. The proportion of type IIB muscle fibers is 55.19%, 57.97%, and 62.37%, respectively. The proportion of oxidized muscle fibers in the longissimus dorsi was higher than that in the triceps brachii and biceps femoris. Moreover, the proportion of different types of muscle fibers in the three parts follows the order: type IIB > type IIA > type I¹². The functional characteristics of each fiber type and their distribution in mutton are detailed in Table 1 and Fig. 1.

Key pathways of mutton flavor formation

Flavor is one of the most important quality characteristics of meat products and a key factor in the overall acceptability of meat products, including fragrance, taste and odor. And also encompassing various taste sensations (sour, sweet, bitter, umami, salty), olfactory perceptions, tactile experiences, and fragrant aroma (fat aroma, barbecue aroma, etc.), collectively contributing to the meat flavor richness^{13,14}. The flavor precursors in meat produce a variety of flavor substances through a series of flavor reactions. Among them, lipid oxidation, Maillard reaction, thiamine degradation, rumen microbial metabolism and the interaction between these reaction products play a key role in the formation of flavor compounds, showing the unique flavor of mutton. The formation mechanism of these reactions illustrated in Fig. 2.

Lipid oxidation

Lipid oxidation is one of the key pathways for the formation of volatile flavor compounds in mutton. Mutton has a relatively high content of unsaturated fatty acids (UFAs), which are structurally unstable. Lipids in mutton are

prone to oxidation during aging, processing, and storage, producing volatile carbonyl compounds with low flavor thresholds, such as aldehydes, ketones, esters, and alcohols, which play a decisive role in mutton aroma¹⁵. However, branched-chain unsaturated fatty acids produced by hydrolysis, such as 4-methyloctanoic acid (MOA), 4-ethyloctanoic acid (EOA) and 4-methylnonanoic acid (MNA), play a decisive role in the odour of mutton¹⁶. The study measured the odor activity values (OAV) of 4-alkyl branched-chain fatty acids in sheep adipose tissue and found that MOA and EOA exhibited OAV values ranging from 4.7–26.5 and 3.1–28.3, respectively, while MNA showed the lowest OAV value (0.0–0.1)¹⁷. In addition, excessive lipid oxidation can lead to warmed-over flavor (WOF), primarily caused by the catalysis of phospholipid oxidation in muscle by released heme and non-heme iron during cooking, ultimately generating undesirable flavor compounds¹⁸.

Maillard reaction

The Maillard reaction refers to a complex series of non-enzymatic browning reactions between reducing sugars (e.g., glucose, ribose) and free amino acids (e.g., amino acids, proteins, peptides) during heating. These reactions involve dehydration, degradation, condensation, and polymerization, ultimately producing compounds like pyrrole, pyrazine, furan, and thiazole, which give mutton a unique flavor and color¹⁹. Glutamic acid, aspartic acid, glycine, alanine and other umami amino acids can form umami compounds through the Maillard reaction, enhancing the saltiness of mutton²⁰. The Maillard reaction can also interact with lipid thermal degradation reactions, influencing the overall flavor.

Thiamine degradation

Thiamine, also known as vitamin B1, is a bicyclic compound containing sulfur and nitrogen. As an important precursor of meat odor, it readily undergoes thermal degradation under neutral or alkaline conditions, generating various nitrogen- or sulfur-containing volatile flavor compounds, such as aliphatic sulfides, thiophenes, and thiacyclic compounds^{21,22}. Research has shown that the content of thiamine in mutton ranging from 0.17 to 0.18 mg/100 g, and its degradation efficiency is easily affected by factors such as pH, temperature, time, and matrix composition²³. Optimizing cooking conditions to enhance the degradation of thiamine can improve the aroma and umami of mutton.

Rumen microbial metabolism

The rumen microbial of ruminants is complex. The vast number of microorganisms present in the rumen significantly affects feed digestion and utilization, energy conversion, fat deposition, and the formation of flavor compounds in meat. For example, rumen yeasts and lactic acid bacteria can break down carbohydrates and proteins in feed, producing volatile compounds like alcohols, esters, and acids. These compounds interact with fat oxidation or Maillard reaction products to form complex flavor systems. In addition, under the action of rumen microbial, tryptophan and tyrosine in the body undergo deamination and decarboxylation reactions, respectively, producing 3-methylindole and 4-methylphenol. 3-Methylindole (skatole) and 4-methylphenol (p-cresol) are also the primary contributors to the

Table 1 | Functional characteristics of muscle fiber types

Muscle fiber types	Main marker	Diameter	Contraction speed	Mitochondrial density	Metabolic type	Myoglobin content	Glycogen content	Fatigue resistance	Distribution in mutton
Type I	MyHC I	Thinnest	Slowest	Highest	Aerobic oxidation	Highest	Lowest	Highest	<i>Biceps femoris</i>
Type IIA	MyHC IIA	Moderate	Moderate	High	Oxidative-Glycolysis	High	Moderate	Moderate	<i>Longissimus dorsi</i> ¹¹
Type IIX	MyHC IIX	Thick	Fast	Moderate	Anaerobic glycolysis	Moderate	High	Low	–
Type IIB	MyHC IIB	Thickest	Fastest	Lowest	Anaerobic glycolysis	Lowest	Highest	Lowest	<i>Longissimus dorsi</i> ¹²

distinctive mutton odor²⁴. 3-Methylindole is closely associated with the characteristic flavor profile of meat sheep raised under grazing conditions. Studies have shown that when the concentration of 3-methylindole exceeds the metabolic threshold in the liver, it enters systemic bloodstream and accumulates in the mutton fat tissue, thereby influencing the mutton odor²⁵. The formation of 4-methylphenol primarily depends on the feed consumed by ruminants and is also strongly correlated with grazing activities and the pH value of the meat²⁶. Its content is positively correlated with grazing time and negatively correlated with the final pH value in meat²⁷. Additionally, research reported that using active components from *Allium* plants and their extracts can inhibit rumen microbial biohydrogenation, thereby suppressing the synthesis of 4-alkyl-branched fatty acids and reducing mutton odor¹.

Regulatory mechanisms of muscle fiber types on meat quality and eating flavor of mutton

The aforementioned four chemical pathways collectively determine the formation of mutton flavor. These pathways are closely associated with the metabolic characteristics of muscle fibers, energy supply patterns, myoglobin content, phospholipid distribution, and IMF deposition. As the fundamental functional unit of muscle tissue, the type composition and metabolic characteristics of muscle fibers directly or indirectly influence the pathways of mutton flavor formation, thereby determining the sensory quality, nutritional value, and flavor profile. Furthermore, the muscle fibers characteristics serve as the intrinsic determinant of meat quality and flavor. Meat quality is closely related to the appearance, palatability, and nutritional value, primarily encompassing meat color, pH value, tenderness, IMF content, juiciness, and flavor. The relative composition of muscle fiber types determines the overall metabolic processes of livestock and poultry muscles post-slaughter, ultimately influencing meat quality. The regulatory mechanisms of muscle fiber types on various indicators are detailed in Table 2.

Effects of muscle fiber types on meat color

Color is one of the intuitive basis for consumers to judge whether the quality of mutton is fresh, which determines the final evaluation and acceptance of consumers. Normal mutton color is bright red or brick red, with myoglobin being the main pigment influencing fresh meat color²⁸. Differences in myoglobin content within muscles directly influence the catalytic activity of heme iron in lipid oxidation, ultimately affecting meat color²⁹. Type I fibers contain approximately four times more myoglobin than type IIB fibers. Muscles rich in type I fibers exhibit a redder color (higher a^* value) and are generally preferred by consumers³⁰. Type IIB fibers contain more glycolytic enzymes and sarcoplasmic reticulum. Post-slaughter anaerobic metabolites from these fibers may lead to PSE meat characteristics, characterized by pale color (Pale), soft texture (Soft), and exudation (Exudative), typically associated with higher lightness (L) values³¹. In Sunit sheep, the proportion of type I fibers in the biceps femoris (16.3%) was significantly higher than in the

longissimus dorsi (8.7%). Correspondingly, the a^* value of the biceps femoris was significantly higher, and the L^* value was significantly lower than those of the longissimus dorsi¹¹.

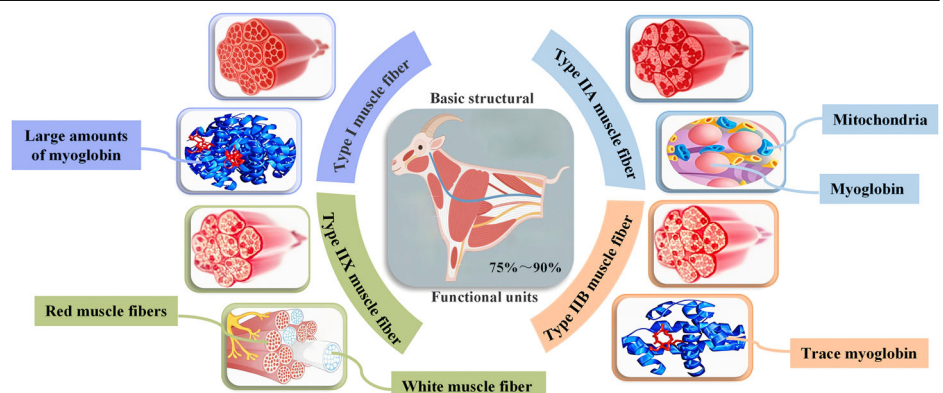
Effect of muscle fiber type on mutton pH

pH is an important indicator of mutton freshness and quality. In general, post-slaughter muscle pH declines more slowly in ruminants than in pork and poultry, attributed to the higher proportion of slow-oxidizing muscle fibers in ruminants³². Moreover, the rapid decline of pH will not only destroy the combination of water and protein, but also affect the protein characteristics of muscle tissue, which will adversely affect the color, water holding capacity, tenderness and other meat quality indicators³³. Studies have shown that an increased proportion of type I fibers can delay the rate of pH decline while improving water-holding capacity, juiciness, and flavor of mutton. Because type I (slow-oxidative) fibers are characterized by high mitochondrial density, strong aerobic metabolism, low glycogen content, and low glycolytic enzyme activity. After slaughter, oxygen depletion inhibits oxidative phosphorylation, but the low glycogen reserve and weak glycolysis capacity of type I fibers lead to less lactate accumulation, thereby slowing the rate of postmortem pH decline³⁴. In contrast, muscles rich in type IIB fibers, due to their high glycogen content and rapid anaerobic metabolic rate, lead to a rapid accumulation of lactic acid and phosphates post-slaughter. This results in a significant decrease in pH and water-holding capacity, along with substantial myoglobin denaturation and loss, which in severe cases may contribute to the formation of PSE meat³⁵. A correlation analysis between meat quality and fiber typing in 15 sheep muscle revealed that in fast-contracting glycolytic muscles, larger fast-contracting glycolytic fibers and lower oxidative activity were typically associated with a lower final pH, whereas in slow-contracting oxidized muscles, higher oxidative activity correlated with elevated 24-h pH and the better water-holding capacity³⁶. Reducing the percentage of type IIB and IIX fibers in ovine longissimus dorsi can decrease the extent of post-slaughter pH decline, thereby improving meat quality³⁷. The rate of pH decrease after cooking affects the optimal pH for the Maillard reaction, thereby influencing meat quality and flavor characteristics. Studies have shown that pH value influences the reaction activity between amino acids and reducing sugars, thereby regulating the rate of the Maillard reaction and the formation of flavor compounds. The reaction rate is lower when pH falls below 6.0³⁸. Specifically, a pH of ~5.5 has been identified as the optimal pH for the formation of volatile compounds in meat aroma³⁹. Pre-slaughter stress can also deplete muscle glycogen, leading to a slow pH decline and the formation of dark, firm, and dry (DFD) meat, negatively impacting flavor⁴⁰.

Effect of muscle fiber type on mutton tenderness

Tenderness refers to the degree to which meat is easily chewed when eaten. It is an important sensory characteristic of mutton quality and directly affects the consumer's eating experience. The shear force is commonly used to evaluate the tenderness of meat. The greater the shear force, the worse the

Fig. 1 | Classification characteristics of four types of muscle fibers in mutton.



tenderness. Muscle fiber types influence tenderness through their diameter and density. Studies have shown that the finer the muscle fibers between the muscle bundles and the greater the density, the higher the deposition ability of IMF than the varieties with thick muscle fibers and low density, and the higher the tenderness of the meat and the juiciness⁴¹. Type I fibers have a smaller diameter and higher density; an increased proportion of these fibers reduces shear force, enhancing tenderness⁴². Conversely, type IIB fibers have a relatively larger diameter, which can reduce tenderness⁴³. Studies also show that in yak meat, the proportion of type I fibers is negatively correlated with shear force, meaning higher type I fiber proportion is associated with lower shear force and better tenderness⁴⁴. And increase in the number of type I muscle fibers will make beef more juicy and delicious, while the number of type IIB muscle fibers will increase the hardness of meat⁴⁵. Resveratrol has been shown to promote the transformation of goat muscle fibers from fast-glycolytic to slow-oxidative types, reducing drip loss and shear force, thereby improving tenderness⁴⁶. Double-muscling in sheep, while increasing meat yield and economic benefits, enlarges fiber diameter, subsequently leading to reduced tenderness and juiciness⁴⁷. Furthermore, after Callipyge mutates into the double-muscling sheep, the proportion of fast-contracting glycolytic muscle fibers (types IIB) increases, while the proportion of slow-contracting oxidized muscles (type I) decreases significantly⁴⁸.

Effects of muscle fiber types on mutton intramuscular fat

Intramuscular Fat (IMF) refers to fat located within the muscle and between muscle fibers. It significantly influences meat tenderness, juiciness, flavor, and visual appearance⁴⁹. When the IMF content is between 2% and 3%, the taste of mutton is the best. And in a certain range, the higher the IMF content, the lower the shear force, the better the water retention performance, the finer the meat taste, and the better the quality⁵⁰. It has been reported that IMF content is negatively correlated with the cross-sectional area of muscle fibers and the proportion of type IIB muscle fibers, but

positively correlated with the proportion of type I muscle fibers⁵¹. The reason is that type I muscle fibers contain more neutral lipids, while type IIB muscle fibers contain less neutral lipids. At the same time, the type I muscle fiber is also rich in phospholipids, which is positively correlated with the juiciness of the meat, which plays an important role in the formation of mutton flavor during cooking⁵². Studies have also confirmed that the proportion of type I muscle fibers in beef is positively correlated with the amount of IMF deposition. The study also found that pig breeds with high IMF content produce more volatile organic compounds during oxidation due to their high content of UFAs in triglycerides, which gives the meat a richer flavor⁵³. However, some scholars believe that there is no universal relationship between IMF and muscle fiber type composition. The two characteristics are relatively independent and can be manipulated separately⁵⁴.

Effect of muscle fiber type on mutton flavor

The metabolic characteristics of muscle fibers have an important influence on meat flavor, which plays a role by affecting the formation and transformation of flavor substances. Oxidized muscle fiber contains more mitochondria and myoglobin, which is conducive to the β -oxidation of fatty acids and the formation of some flavor precursors, such as aldehydes, ketones and other volatile flavor substances. The increase of such muscle fiber content can improve the umami and flavor richness of muscle. On the contrary, the increase in the proportion of glycolytic muscle fibers may lead to the accumulation of lactic acid, affect the pH value and water holding capacity of the meat, and increase the sour and astringent taste of the meat⁵⁵. Besides type I fiber proportion shows positive correlations with the content of essential amino acids like lysine, valine, and glutamic acid, thereby enhancing nutritional quality and flavor precursor levels. In contrast, type IIB fiber proportion shows negative correlations with amino acid content⁵⁶. In the flavor study of Jinhua pork skeletal muscle confirmed that muscles predominantly composed of type I and type IIA (oxidized) muscle fibers are

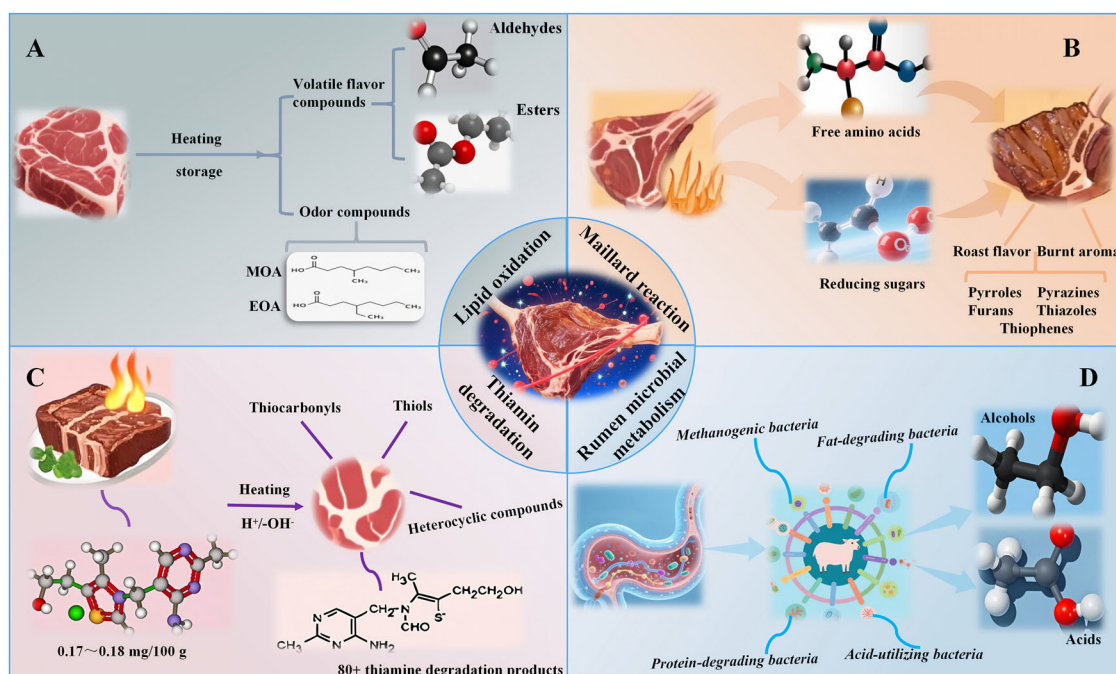


Fig. 2 | The formation mechanism of mutton flavor. **A** Mutton undergoes lipid oxidation during heating or storage to produce volatile flavor compounds (e.g., aldehydes, esters, etc.) and odor compounds (e.g., MOA, EOA). **B** Heating can stimulate the Maillard reaction of free amino acids and reducing sugars in mutton, producing pyrroles, pyrazines, furans and other compounds with roasted and burnt flavor. **C** The thiamine in mutton undergoes thermal degradation under neutral or alkaline conditions, producing volatile aroma compounds containing nitrogen or sulfur (e.g., thiocarbonyls, thiols, heterocyclic compounds, etc.). **D** There are various

microorganisms in the rumen of mutton (e.g., fat-degrading bacteria, protein-degrading bacteria, acid-utilizing bacteria, etc.), which can react with feed, energy, fat etc. to produce substances such as alcohols, esters, and acids that contribute significantly to the flavor of mutton.

Table 2 | Regulatory mechanisms of muscle fiber types on meat quality and eating flavor of mutton

Index	Muscle fiber type	Key mechanism/characteristic	Impact on quality and flavor
Meat color	Type I (Slow-oxidative)	High myoglobin content (~4x of type IIB) higher a* value (redder color) ³⁰	Positive: brighter red color, consumer preferred
	Type IIB (Fast-glycolytic)	High glycolytic enzyme content, rapid post-slaughter glycolysis associated with PSE meat ³¹	Negative: pale color, soft texture, exudation, reduced acceptance
pH value	Type I (Slow-oxidative)	High mitochondrial density, low glycogen, weak glycolytic capacity → slower pH decline better water-holding capacity ³⁴	Positive: better water-holding, juiciness and flavor
	Type IIB (Fast-glycolytic)	High glycogen content, rapid anaerobic metabolism → rapid lactate accumulation → fast pH decline; risk of PSE meat ³⁵	Negative: protein denaturation, drip loss, reduced quality
Tenderness	Type I (Slow-oxidative)	Smaller diameter, higher density reduced shear force higher IMF deposition capacity ^{41,42}	Positive: enhanced tenderness, finer texture
	Type IIB (Fast-glycolytic)	Larger diameter, increased shear force, reduced tenderness ⁴³ ; Callipyge mutation increases type IIB proportion ⁴⁸	Negative: coarser texture, tougher meat
IMF	Type I (Slow-oxidative)	Rich in neutral lipids and phospholipids, positively correlated with IMF deposition enhances juiciness ^{51,52}	Positive: better marbling, juiciness, flavor richness
	Type IIB (Fast-glycolytic)	Low neutral lipid content, negatively correlated with IMF; increased meat hardness ^{51,52}	Negative: lower fat deposition
Flavor	Type I & IIA (Oxidative)	Rich in phospholipids and free amino acids (Glu, Asp, Ala, Gly), promotes lipid oxidation flavor precursors and enhances umami ⁵⁷	Positive: rich flavor, strong umami, nutritional quality
	Type IIB & IIX (Glycolytic)	Positive correlation with MOA, MNA (mutton odor compounds); lower free amino acid levels, insufficient umami intensity ^{56,57}	Negative: strong mutton odor

rich in free amino acids, particularly umami amino acids (aspartic acid, glutamic acid) and sweet amino acids (alanine, glycine), which can enhance the umami flavor and mellow taste of meat. In contrast, muscles dominated by type IIB fibers, although abundant in volatile aldehydes, exhibit lower total free amino acid levels and insufficient umami intensity⁵⁷. Moreover, the study through correlation heatmap analysis between the expression of MyHC subtypes and specific volatile compounds. Found that MyHC IIB and IIX subtypes were positively correlated with MOA and MNA. The two branched chain fatty acid derivatives that are the main substances producing the mutton gamey taste⁵⁶. As a key flavor precursor, phospholipids play a decisive role in meat flavor. The positive effect of type I muscle fiber on flavor may be due to its high phospholipid content, which can produce more flavor substances through thermal degradation reaction. However, polyunsaturated fatty acids in phospholipids also increase the risk of fatty acid oxidation and rancidity, leading to the formation of bad flavor²⁸.

Factors affecting the muscle fiber types, meat quality, and flavor in mutton

Breed and genotype

Muscle fiber characteristics are moderately to highly heritable. Due to differences in genetic background, muscle fiber type composition, distribution proportions, and metabolic properties vary considerably among different sheep breeds. In Sunit sheep, the diameter and cross-sectional area of various fiber types were significantly lower than in Bamei mutton sheep and Small-tailed Han sheep, which partly explains the superior meat quality of Sunit sheep⁵⁸. The fiber diameter in the longissimus dorsi of the Hu sheep was smaller than that of the Dorset sheep, while the fiber density was significantly greater, indicating that the texture of the Hu sheep was finer and the tenderness was better⁵⁹. This reveals that indigenous Chinese breeds (e.g., Hu sheep, Sunit sheep) finer typically have finer fibers and higher oxidative potential compared to introduced meat breeds (e.g., Dorset), which is related to their superior meat quality traits. Regarding flavor, compared to goat meat, sheep meat is generally softer, more tender, and juicier, with a higher IMF content, a relatively milder odor, and better edible value⁶⁰. Kazakh sheep meat is rich in protein, total essential amino acids, and umami amino acids, with favorable flavor⁶¹, while Ujimqin sheep meat has higher IMF content, a reasonable and diverse fatty acid composition, and a

rich flavor⁶². However, in breeding programs, focusing solely on single traits like “meat yield” is insufficient. Setting a threshold for the ratio of type I to type IIB fibers could serve as a selection criterion for excellent mutton breeds. This approach can maintain a certain growth rate while leveraging the high phospholipid content of oxidative fibers to promote lipid oxidation and obtain more aromatic compounds, while avoiding the deterioration of tenderness associated with excessive glycolytic fibers.

On the other hand, genetic selection for favorable genes can improve meat quality and flavor. Muscle fiber types show moderate to high heritability in sheep, pigs, and cattle. These features can be incorporated into genome selection programs to enhance the muscle fiber profiles and meat quality traits in sheep, promoting high-quality development of the mutton industry⁵⁸. Research has found that increased FoxO1 gene expression is negatively correlated with MyHC I mRNA levels and positively correlated with MyHC IIX and MyHC IIB levels, identifying it as a core regulator of fiber type transformation⁶³. The Halothane (Hal) gene can downregulate the expression of type IIA fiber gene and upregulate the expression of type IIX fiber, increase the proportion of white (glycolytic) fibers, and reduce meat quality. The Rendement Napole (RN) gene mutation can decrease MyHC IIB mRNA and promote IIA and IIX mRNA expression. Mutations in these two genes altered the composition of muscle fiber type, ultimately having anegative impact on meat quality traits⁶⁴. Furthermore, genes such as Lipoprotein Lipase (LPL)⁶⁵, Doublecortin-Like Kinase 1 (DCLK1)⁶⁶, and Lipin-1 (LPIN1)⁶⁷ can influence IMF deposition, indirectly affecting muscle fiber type and flavor compound accumulation. These can serve as molecular markers for assisted selection in the genetic improvement of mutton quality traits, accelerating industry development.

Age and sex

Age can change the direction of muscle fiber type transformation to affect meat quality. The proportion of oxidized fiber is negatively correlated with age, while the proportion of glycolytic fibers is positively correlated. The number of muscle fibers is largely fixed at birth, but the fiber types can switch in order from I → IIA → IIX → IIB. As age increases, the proportion of oxidized fibers gradually decreases, while the proportion of glycolytic fibers increases. Fibers generally become thicker, and have an increased cross-sectional area, which may lead to tougher texture and reduced tenderness⁶⁸.

Studies report that the Longissimus dorsi of 7-day-old Bamei and Landrace pigs was rich in Type IIA and IIX fibers, but by 180 days, these proportions significantly decreased, while the proportion of type IIB fiber significantly increased, and the change in Landrace pigs are more pronounced⁶⁹. Both Xinjiang brown cattle and Angus cattle showed a trend of decreasing type I fiber density with age (from 3 to 24 months) during the growth stage⁷⁰. Age can also affect the deposition of IMF in muscles, as well as the accumulation of chemical compositions and flavor compounds. Meat from young mutton is tender, with fine fibers and low connective tissue content, but flavor may be less developed. As age increase, mutton fibers coarsen, fat deposition increases, fatty acid composition changes, and the levels of some flavor precursors change, leading to a decrease in tenderness and a more pronounced flavor⁷¹. In Sunit sheep, the content of polyunsaturated fatty acid (PUFA) in muscle significantly decreased with age, while the content of saturated fatty acid (SFA) and stearic acid gradually increased, leading to an increase in the accumulation of mutton odor compounds⁷².

Gender affects the quality, flavor, muscle fiber characteristics, and chemical composition of mutton through hormone levels. Generally, rams grow faster have more developed muscle, but their meat tends to be coarser, with larger fiber diameter and stronger odor. This is because the higher levels of androgen in rams promote fat metabolism, alter the content and composition of short-chain and unsaturated fatty acids in fat, and enhance meat odor. Ewe meat is relatively tender, with the smallest fiber diameter and milder odor⁷³. Castrated male sheep (wethers) have increased levels of follicle-stimulating hormone (FSH) and luteinizing hormone (LH), and decreased levels of gonadotropins and the male hormone testosterone. This hormonal change effectively reduces the synthesis of branched-chain fatty acids associated with mutton odor, lowers meat odor, and significantly improves flavor⁷⁴. In agricultural practices, raising castrated male sheep is recommended. If balancing reproduction and meat quality is needed, ewes are a secondary option, but direct fattening of intact rams should be avoided.

Feeding methods

Ruminant feeding systems primarily include natural grazing, house feeding, and supplementary feeding grazing. Under these systems, the flavor profiles of animal products and muscle fiber types differ. Grazing increases animal movement and promotes the development of oxidative (type I) fibers. In addition, PUFA from pasture can be incorporated into muscle, resulting in a more natural and complex mutton flavor⁷⁵. Conversely, confinement systems may increase the proportion of glycolytic (type IIB) fibers while accelerating weight gain. High-grain diets in confinement can also produce more propionic acid in the rumen, leading to increased deposition of branched-chain fatty acids associated with mutton odor⁵. Studies also showed that compared with confinement-fed sheep, grazing sheep have the finest fibers, significantly higher MyHC I expression, and significantly lower MyHC IIA and MyHC IIB expression. The shear force values and certain color parameters (L^* and b^*) were also significantly decreased in grazing sheep⁷⁶. When Sunit sheep were raised under three different feeding regimes, the confinement group had the highest proportion of type IIB fibers, while the grazing and supplementary grazing groups had higher proportions of type I and IIA fibers, resulting in tender meat²⁸, indicating grazing can enhance mutton quality and flavor. However, grazing systems have longer finishing cycles, resulting in lighter carcasses in the short term, making it difficult to meet the premium mutton standards quickly. They also bring greater pressure to grassland resources and lower economic efficiency. In order to accelerate fattening, a combination system of daytime grazing with morning and evening supplemental feeding is often adopted. This model not only shortens the finishing cycle and saves feeding costs but also meets daily nutritional needs, producing high-quality mutton⁷⁷.

Dietary nutrition

Nutrition is crucial for muscle fiber growth. Dietary energy, protein levels, and additives can influence fiber type composition by modulating muscle fiber metabolism. Studies feeding Hu sheep with different energy level diets found that reducing dietary energy level decreased fiber area, increased fiber

density, and improved meat quality⁷⁸. However, other research suggests appropriately increasing dietary energy levels in pigs can increase the proportion of oxidative fibers, thereby improving meat quality⁷⁹. Supplementing the diet of Duhan crossbred sheep with 100 mg/kg fermented wheat bran polysaccharides significantly increased the expression of oxidative fiber-related genes in the Longissimus dorsi, decreased the expression of intermediate fiber genes and shear force values, improving tenderness⁸⁰. Adding different doses of allicin to the diet of Guizhou black goats found that allicin promoted the transformation of fiber types from type II to type I, improved meat quality by regulating relevant enzyme activities and gene expression, and also altered fatty acid composition and amino acid profiles in mutton, thereby affecting the generation of volatile flavor compounds⁸¹. Furthermore, dietary components directly influence flavor compound accumulation. Adding 3% compound Chinese herbal medicine reduced EOA and MNA content in mutton, lessening odor⁸². Feeding sainfoin rich in tannins not only inhibits partial oxidation of fatty acids in the rumen, increasing n-3 PUFA content, but also minimizes unpleasant odors from forage, improving mutton flavor⁸³. In practice, combined strategies like “low-energy restricted feeding plus antioxidant supplementation” can be employed to slow the rate of fiber conversion to glycolytic types and improve meat quality and flavor.

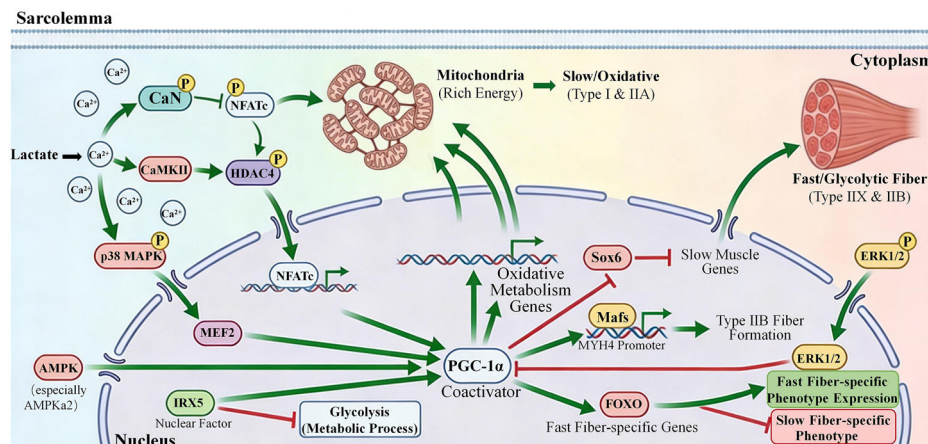
Other key factors

Heat, transportation, pre-slaughter stress, and post-slaughter handling can all affect the muscle fiber structure and flavor compound release of mutton. Heat stress (HS) can accelerate the conversion of oxidative muscle fibers (type I) to glycolytic muscle fibers (type II), thereby increasing anaerobic metabolism in muscle. This shift elevates the rate of lactate accumulation post-slaughter, accelerates muscle pH decline, and results in tougher meat with reduced water-holding capacity. Furthermore, HS activates the hypothalamic-pituitary-adrenal (HPA) axis and increases cortisol secretion. Cortisol inhibits muscle protein synthesis and promotes protein degradation; increased protein degradation further reduces the levels of umami amino acids such as glutamic acid and aspartic acid, leading to a decrease in high-quality flavor compounds⁸⁴. Transportation stress is a significant source of pre-slaughter stress, which promotes glycogenolysis and anaerobic glycolysis by activating the AMPK signaling pathway. In Hu sheep subjected to 8 h of long-distance transportation, muscle glycogen content was significantly decreased on day 0, while the activities of lactate dehydrogenase (LDH) and hexokinase (HK) were markedly elevated, indicating that transportation stress accelerates glycolytic metabolism in muscle fibers⁸⁵. Depletion of muscle glycogen leads to an abnormal rate of post-slaughter pH decline, which may result in the occurrence of PSE or DFD meat. In addition, transportation stress increases the shear force value of mutton, reduces meat tenderness, and decreases IMF content, leading to a drier texture and inferior flavor⁸⁶. Goats transported for 6 h during the hot season exhibited higher pH value, drip losses, and shear force, while significantly shortening sarcomere length and deteriorating meat quality⁸⁷. Pre-slaughter stress can be divided into long-term chronic stress and short-term acute stress, both of which can affect the muscle fiber types and enzyme activity of mutton. Oxidative muscle fibers (type I) are rich in succinate dehydrogenase (SDH) and cytochrome oxidase, whereas glycolytic muscle fibers (type IIB) are abundant in LDH and phosphorylase. Under stress conditions, LDH activity in type IIB fibers significantly increases, accelerating the production of lactic acid⁸⁶. Studies on the intervention of α -lipoic acid have demonstrated that the AMPK-SIRT1-PGC-1 α signaling pathway can promote the transformation of muscle fibers from type IIB to type IIA, reduce LDH activity, and enhance SDH activity, thereby improving meat quality⁸⁸. This indicates that pre-slaughter stress management can improve meat quality by regulating enzyme activities related to muscle fiber types. In addition, the duration of lairage is crucial for relieving animal stress. Research have shown that muscle glycogen levels in Small-tailed Han sheep partially recover after 24 h of lairage following transport. However, excessive grazing time (exceeding 12 h) may lead to the accumulation of additional stress⁸⁹. Regarding the post-slaughter processing of mutton, such as chilling method

Table 3 | Factors affecting muscle fiber types, meat quality, and flavor in mutton

Influencing factor	Specific condition	Effect on muscle fiber types	Impact on meat quality and flavor	Practical application
Breed/geno-type	Indigenous vs introduced breeds	Indigenous breeds have smaller fiber diameter, higher oxidative potential and better meat quality ⁵⁹	Indigenous breeds: superior tenderness, flavor, lower odor	Select indigenous breeds for premium quality; use I/IB ratio as selection criterion
	Gene expression and gene mutations	FoxO1: negatively correlated with MyHC I positively correlated with MyHC IIX and IIB ⁶³ Hal: increases type IIX RN: decreases MyHC IIB, promotes IIA and IIX ⁶⁴	Alters fiber type composition toward glycolytic phenotype Negative impact on meat quality traits	Molecular marker for assisted selection in breeding programs Avoid unfavorable genotypes in breeding programs
	Age	Young vs Adult Type I fiber density decreases, fibers become thicker, type IIB increases significantly with age ⁶⁵	Decreased tenderness; more pronounced flavor; coarser texture ⁶⁶	Optimize slaughter age for desired quality (typically younger for tenderness)
Sex	Rams	Larger fiber diameter, more developed muscle, higher glycolytic fiber proportion ¹⁰⁷	Coarser meat, stronger odor due to higher androgen levels ¹⁰⁸	Avoid direct fattening of intact rams for quality meat production
	Ewes	Smallest fiber diameter, mildest odor, better tenderness ⁷³	Tender meat, mild odor	Mainly used for breeding and production
	Wethers	Increased FSH/LH, decreased testosterone; reduced branched-chain fatty acids ⁷⁴	Reduced mutton odor, improved flavor ¹⁰⁹	Mainly used for fattening
Feeding methods	Grazing	Higher MyHC I expression, lower MyHC IIA/IIB; finer fibers ⁷⁶	More natural flavor; higher PUFA ⁷⁵	Premium quality production; suitable for grassland areas
	Confinement	Higher type IIB proportion; accelerated weight gain; more propionic acid ⁷⁵	Increased mutton odor (branched-chain fatty acids); faster growth	Commercial production; requires odor mitigation strategies
	Supplementary feeding	Higher type I and IIA proportions; tender meat ¹¹⁰	Shortened finishing cycle; high-quality mutton production ¹¹¹	Cost-effective with acceptable quality
Dietary nutrition	Energy restriction	Decreased fiber area, increased fiber density, improved meat quality ⁷⁸	Improved meat quality through fiber density increase	Strategy for improving meat quality in feedlot systems
	Feed additive	Fermented wheat bran polysaccharides: oxidative fiber ↑, intermediate fiber ↓ ⁸⁰ Allicin supplementation: Promotes Type II → Type I transformation; improves meat quality ⁴¹	Reduced shear force; improved tenderness ⁸³ Altered fatty acid and amino acid profiles; reduced odor compounds ⁸¹	Functional feed additive for quality improvement Natural feed additive for fiber type modulation and flavor improvement
	Heat stress/transportation stress	Heat stress: Accelerates Type I → Type II conversion; increases anaerobic metabolism ¹¹² Transportation stress: Activates AMPK pathway; accelerates glycolytic metabolism; glycogen depletion ⁸⁵	Reduced water-holding capacity; tougher meat; decreased umami amino acids ⁸⁴ Increased shear force; reduced tenderness; lower IMF; PSE/DFD risk ⁸⁶	Implement cooling and stress reduction measures pre-slaughter, and optimize transportation duration
Other factors	Post-slaughter aging	Myofibrillar protein degradation; improved tenderness; flavor precursor release ⁹¹	Stabilized protein solubility, full flavor precursor release ⁹²	Control aging conditions (temperature, time) for optimal quality

Fig. 3 | Molecular regulatory mechanisms underlying muscle fiber type transformation. The left panel represents pathways promoting the formation of slow-twitch/oxidative fibers (Type I & IIA), whereas the right panel depicts pathways driving fast-twitch/glycolytic fiber (Type IIX & IIB) development. Green arrows denote activation or promotion, while red arrows indicate inhibition or blockade. Signals are transduced from the cytoplasm to the nucleus, ultimately regulating gene expression.



and aging time, research has indicated that ultra-rapid chilling can delay glycolysis rate and improve tenderness by increasing the phosphorylation levels of aldolase A (ALDOA), phosphofructokinase (PFKM), and phosphoglycerate kinase (PGK), while inhibiting their acetylation levels⁹⁰. However, inappropriate rapid chilling may lead to excessive contraction of muscle fibers, shortened sarcomere length, and increased meat hardness. Post-slaughter aging is a key process for the degradation of myofibrillar structural proteins and the formation of flavor compounds. Myofibrillar proteins are degraded by endogenous enzymes such as calpains, cathepsins, and proteasomes, leading to muscle fiber fragmentation, increased myofibrillar fragmentation index (MFI), and improved tenderness⁹¹. Studies have found that the period of 24–48 h post-slaughter in goats represents a superior stage for meat tenderization, during which changes in myofibrillar protein solubility stabilize, and flavor precursors are fully released⁹². The effects of various intrinsic and extrinsic factors on muscle fiber types and their consequent impacts on meat quality and flavor are comprehensively summarized in Table 3.

Molecular regulation of muscle fiber type transformation

Different muscle fiber types exhibit significant differences in metabolic characteristics, contractile function, and energy supply, directly influencing meat quality traits such as tenderness, color, and flavor. The transformation of muscle fiber types is regulated by a complex molecular network. This network involves the coordinated interplay of multiple transcription factors and signaling pathways that collectively determine the directional differentiation of muscle fibers toward oxidative (type I & IIA) or glycolytic (type IIX & IIB) phenotypes. The underlying regulatory mechanisms are detailed in Fig. 3 and Table 4.

Transcription factor regulatory network

Transcription factors regulating muscle fiber types mainly include peroxisome proliferator-activated receptor γ coactivator 1 α (PGC-1 α), Sox 6, Mafk, IRX5, etc. PGC-1 α plays an important role in regulating mitochondrial synthesis, energy substrate utilization and oxidative metabolism during muscle fiber formation⁹³. It was found that the activity of genes related to mitochondrial oxidative metabolism in type II muscle fibers of mice was significantly enhanced after overexpression of PGC-1 α in transgenic mice driven by muscle creatine kinase (MCK) gene promoter⁹⁴. In addition, by constructing a PGC-1 α -deficient mouse model, we observed a significant change in the type of skeletal muscle fibers in mice. The proportion of oxidized fibers in the muscle decreased, while the proportion of glycolytic fibers increased. These findings confirm that PGC-1 α is a key factor in regulating muscle fiber type conversion and can induce the transition from fast muscle fibers to slow muscle fibers⁹⁵. In humans, the expression of PGC-1 α was positively correlated with oxidized muscle fibers (type I) and

negatively correlated with glycolytic muscle fibers (type IIB)⁹⁶. Sox 6 is enriched in the expression of fast muscle fiber genes, and promotes the transformation of slow muscle to fast muscle by inhibiting slow muscle genes⁹⁷. Transcription factor Mafk can specifically bind to the MYH4 promoter to increase the formation of type IIB muscle fibers⁹⁸. The study also found that IRX5 can inhibit glycolysis and activate PGC-1 α expression, thereby promoting slow muscle differentiation⁹⁹.

Signaling pathway regulation

The main signaling pathways regulating the transformation of muscle fiber types include calcium signaling pathway, FOXO signaling pathway, MAPK-related pathway, PI3K/AKT pathway, etc. The Ca²⁺ signaling pathway includes calcineurin (Ca N) and calcium-dependent protein kinase (Ca MK). Lactic acid is a muscle factor. In the Ca N signaling pathway, lactic acid can increase the concentration of intracellular Ca²⁺, promote the dephosphorylation of nuclear factor of activated T-cells cytoplasmic (NFATc) in myotubes, activate the Ca N/NFATc1 signaling pathway, increase muscle endurance, and induce the conversion of muscle fiber types to slow muscle fibers¹⁰⁰. The increase of Ca²⁺ can lead to the increase of p38 MAPK phosphorylation level in PPAR γ overexpression pigs, which can stimulate MEF 2 and PGC-1 α to affect muscle fiber type and oxidation ability, and promote the transformation of muscle fibers from fast muscle to slow muscle¹⁰¹. In the Ca MK signaling pathway, the Ca MK signaling pathway promotes the phosphorylation of histone deacetylase 4 (HDAC4) by activating CaMKII kinase, significantly increases the nuclear expression of HDAC4, and promotes the transformation of muscle fiber type from fast muscle to slow muscle¹⁰². FOXO signaling pathway is related to muscle fiber type and can promote the expression of fast muscle fiber related genes. Knockout of foxo gene in aged mice can significantly increase the diameter of soleus muscle fibers¹⁰³. Overexpression of PKA catalytic subunit can induce muscle fiber transition from fast muscle to slow muscle by inhibiting the activity of foxo¹⁰⁴. Adenosine monophosphate-activated protein kinase (AMPK) is a serine/threonine protein kinase widely present in eukaryotic cells, serving as a key regulator of intracellular energy metabolism. It primarily includes extracellular regulated protein kinases (ERK), c-Jun N-terminal kinase (JNK), and p38 MAPK¹⁰⁵, which can regulate fiber type transformation through their respective signaling pathways. The signal transduction of ERK1/2 can promote the expression of fast muscle fiber phenotype and inhibit the expression of slow muscle fiber phenotype. It is further confirmed that phosphorylated ERK1/2 can promote the transformation of muscle fiber type from slow muscle fiber to fast muscle fiber¹⁰⁶. Research has revealed that in the biceps femoris muscle of Mongolian sheep, AMPK activity along with the expression of AMPK α 2 and PGC-1 α genes can promote the transformation of muscle fiber types from MyHC IIX, MyHC IIA, and MyHC IIB to MyHC I. This shift alters muscle fiber composition and ultimately influences meat quality⁷⁶. These findings

Table 4 | Molecular regulation of muscle fiber type transformation

Regulatory category	Molecular regulator	Regulatory mechanism	Fiber type transformation direction	Downstream targets	Functional consequence for meat quality	Potential application in mutton production
Transcription Factors	PGC-1α	Master regulator of mitochondrial biogenesis and oxidative metabolism ⁸³ ; drives slow-twitch fiber formation ⁸⁵	Type II → Type I	↑ Mitochondrial genes, oxidative enzymes (SDH, COX) ^{88,113}	Improved tenderness, color stability, flavor richness ¹¹⁴ , enhanced oxidative capacity ¹¹⁵	Key target for genetic selection and nutritional intervention to improve quality
	Sox6	Repressor of slow muscle fiber genes; promotes fast muscle fiber gene expression ⁹⁷	Type I → Type II	↓ MyHC I, slow fiber genes; ↑ MyHC II B, fast fiber genes ¹¹⁶	Potential negative impact on meat quality (increased glycolytic fibers) ³⁴	Marker to avoid in breeding programs; target for inhibition to maintain quality
	Mafs	Binds to MYH4 promoter to enhance type IIB fiber formation and fast-twitch phenotype ⁸⁸	Type I/II A → Type II B	↑ MYH4 expression; ↑ Type II B fiber formation ⁸⁸	May reduce tenderness and increase cooking loss ¹¹⁷	Understanding fast fiber development; manage to prevent excessive glycolytic conversion
	IRX5	Inhibits glycolysis and activates PGC-1α expression; promotes slow muscle differentiation ⁸⁹	Type II → Type I	↓ glycolytic gene expression; enhanced adipogenesis ⁸⁹	Potential candidate genes for meat quality ¹¹⁸	Novel target for promoting oxidative fiber development and quality improvement
Signaling Pathways	Ca ²⁺ /CaV/NFATc1	Lactic acid increases intracellular Ca ²⁺ , promotes NFATc1 dephosphorylation, induces slow fiber conversion ¹⁰⁰	Type II → Type I	↑ Slow fiber endurance; ↑ oxidative metabolism; activated by lactic acid	Enhanced muscle endurance; improved postmortem meat quality (slower pH decline) ¹¹⁹	Lactic acid-based interventions; exercise regimens to promote oxidative fibers
	Ca ²⁺ /CaMKII/HDAC4	CaMKII phosphorylates HDAC4, increases nuclear HDAC4 expression, promotes fast-to-slow fiber transition ¹⁰²	Type II → Type I	↑ Nuclear HDAC4; histone deacetylation; slow fiber gene activation ¹⁰²	Improved oxidative capacity; better color and water-holding capacity ^{101,120}	Target for epigenetic modulation of fiber type; calcium supplementation strategies
	FOXOs	Promotes fast muscle fiber-related gene expression; knockout increases soleus muscle fiber diameter ¹⁰³	Type I → Type II	↑ Fast fiber genes; muscle atrophy prevention; age-related fiber changes ¹⁰³	Potential positive impact on tenderness ²¹	Age-related quality management; FOXO inhibitors for aged animal meat quality
	ERK1/2	Phosphorylated ERK1/2 promotes fast fiber phenotype and inhibits slow fiber phenotype expression ¹⁰⁶	Type I → Type II	↑ Fast fiber phenotype markers; ↓ Slow fiber phenotype markers ¹⁰⁶	Shift toward glycolytic phenotype; faster pH decline; reduced water-holding ²²	Avoid excessive ERK activation; manage stress to prevent fast fiber dominance
	PKA	Induces fiber transition from fast to slow by inhibiting FOXO activity through phosphorylation ¹⁰⁴	Type II → Type I	↓ FOXO transcriptional activity; muscle plasticity regulation ¹⁰⁴	Muscle fiber plasticity; potential for quality improvement through exercise/nutrition ¹²³	Exercise mimetics; PKA activators for fiber type optimization

demonstrate that the AMPK/PGC-1 α signaling pathway regulates postnatal muscle fiber type conversion. Therefore, modulating this signaling pathway represents a potential strategy to increase the proportion of oxidative fibers and thereby enhance meat quality.

Conclusion and prospects

Mutton quality and flavor are core factors restricting consumer acceptance and industrial development, and muscle fiber type composition is the key determinant regulating mutton color, pH, tenderness, juiciness, intramuscular fat deposition, and flavor compound formation. This review systematically clarifies the influence of muscle fiber types on the quality and flavor formation of mutton, revealing the regulatory mechanisms of muscle fiber types on meat quality and flavor formation. And comprehensively summarize the key influencing factors and molecular regulatory networks underlying muscle fiber type transformation, providing theoretical support for the genetic improvement and precise production of high-quality mutton.

Oxidized muscle fibers (Types I and IIA) are finer diameters, higher levels of myoglobin and phospholipid content, which enhance the mutton tenderness, juiciness, and flavor richness. In contrast, the diameter of glycolytic fibers (type IIB) is usually thicker and positively correlated with indicators such as shear force and cooking loss, while showing negative correlations with other meat quality parameters. The increase in the proportion of type II B fibers can also increase the content of special flavor compounds such as MOA, EOA and MNA. In severe cases, may lead to PSE meat. Therefore, type I and type IIA muscle fibers can be targeted as key selection indicators in mutton production. Furthermore, the transformation of muscle fiber types is coordinately regulated by a series of transcription factors and signaling pathways. Targeted intervention in these regulatory networks is crucial for increasing the proportion of oxidized muscle fibers and improving meat quality and flavor.

The continuous development of modern molecular biotechnology has provided a multi-dimensional research method for muscle tissue research. However, current research still has shortcomings. Firstly, the molecular regulatory mechanisms linking muscle fiber types and flavor compounds are not fully elucidated; for instance, the pathways connecting phospholipid metabolism and fiber types require further clarification. Secondly, there is a lack of data supporting the distribution patterns and regulation methods of high marble patterned muscle fiber types in mutton. Thirdly, existing research predominantly focuses on a few high-value commercial cuts like the longissimus dorsi and biceps femoris, lacking a systematic understanding of the dynamic distribution and transformation patterns of fibers in different body parts.

Based on this, future research can focus on three areas. Firstly, utilizing multi-omics technologies to identify key genes and signaling pathways regulating muscle fiber type transformation. Secondly, study the muscle fiber characteristics of high-marbling grade mutton and establish an accurate model between fiber type and flavor quality. Thirdly, develop a comprehensive technical system that integrates “genetic selection + nutritional regulation + management practices.” This system aims to guide fibers toward the ideal types, while identifying molecular markers closely linked to desirable fiber types and low-odor traits. These markers can be used for molecular design breeding of mutton quality to accelerate the development of new breeds. Furthermore, incorporating rapid detection technologies like electronic noses and electronic tongues to establish a more comprehensive evaluation system for mutton quality and flavour, facilitating precise assessment and control of meat attributes.

Data availability

No original experimental data were generated in this review. The data supporting the tables and figures are derived from previously published studies, databases and reports, all of which are cited in the manuscript. The curated information extracted from these sources and used to generate the summary tables and schematic figures. Additional details are available from the corresponding author upon reasonable request.

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Author contributions

M.C.: Writing – review & editing. X.Z.: Writing – original draft and prepared figures. N.Z.: Resources. A.S.G.: Conceptualization and Supervision. F.J.: Conceptualization. X.S.: Methodology. J.C.: Supervision. Q.X.: Funding acquisition, Project administration and Writing – original draft.

Competing interests

The authors declare no competing interests.

Additional information

Correspondence and requests for materials should be addressed to Qi Xiong.

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